

Observation report on Ckoirama and advanced image processing of the galaxy NGC 300: Sensor characterization and Deconvolution

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(Dated: December 16, 2025)

During the night of November 14, 2025, observations of the spiral galaxy NGC 300 were made with the Chakana telescope (CDK24, 0.6 m, f/6.5) at the Ckoirama Observatory. Images were acquired with the FLI ProLine 16801 CCD camera using u, g, r, and i broadband filters. The science exposures consisted of 120-second integrations, supplemented by a full set of calibration frames: bias, dark (120 s), and twilight flat fields. Observing conditions were challenging, characterized by strong winds (up to 8.4 m/s), turbulence, and CCD sensor heating. In addition to standard reduction, this work presents a detailed characterization of the CCD sensor, including calculations of read noise (RON), gain, and dark stream stability. Furthermore, the post-processing process implements the Richardson-Lucy deconvolution algorithm to recover the spatial resolution lost due to atmospheric visibility. The final result includes images with stacked filters and a composite analysis of the sensor's performance.

I. INTRODUCTION

The galaxy NGC 300, located in the constellation Sculptor, is a spiral galaxy similar in morphology to M33. Its proximity and face-on orientation make it an excellent target for resolving stellar populations and studying the structure of the interstellar medium.

This report details the acquisition and processing of NGC 300 imaging data as part of the Master's degree course in Observational Astronomy II. Unlike standard reduction procedures, this exercise emphasizes two critical aspects:

- 1. Sensor Characterization:** A statistical analysis of the calibration frames (Bias, Darks, and Flats) was performed to derive physical parameters of the camera, such as the Gain (e^-/ADU) and the suppression of fixed pattern noise via stacking strategies (Mean vs. Median).
- 2. Advanced Processing:** Implementation of the Richardson-Lucy deconvolution algorithm to enhance the definition of the galactic nucleus and star-forming regions, mitigating the effects of atmospheric turbulence noted during the night.

The following sections describe the observational conditions, the mathematical derivation of sensor properties, the normalization strategy for unstable flat-fields, and the final image reconstruction.

II. METHODOLOGY AND RESULTS

A. Observation

The observation session took place on the night of November 14, 2025, at the Ckoirama Observatory,

operated by the University of Antofagasta. The team consisted of all the students in the course, under the supervision of the faculty.

The instrument used was the Chakana telescope (0.6 m, f/6.5), equipped with the FLI ProLine 16801 CCD camera. The observers arrived at the site at 18:15 hrs local time (21:15 UTC).

1. Conditions and Technical Issues

The weather conditions were marked by significant wind throughout the night. Wind speeds were recorded at 8.4 m/s at the beginning of the session (29°C), decreasing slightly to 5.4 m/s later in the night (16.6°C), but gusts remained a factor causing turbulence. Humidity remained stable and low at approximately 26%.

Technical difficulties were encountered during the session. Notably, the camera disconnected during the acquisition of dark frames (00:28 local time), requiring repairs after cycle #20. Additionally, image #5 in the *u* filter was affected by a satellite trail, and the first science images were discarded due to defocus.

2. Data Acquisition

The acquisition sequence was executed as follows:

- Bias Frames:** A set of 101 bias frames was acquired at 20:00 local time to ensure robust statistical analysis of the readout noise.
- Flat-Field Frames:** Twilight flats were acquired for filters *u, g, r, i* with exposure times ranging from 4s to 5s.

- *u* band: 22 frames (abandoned at cycle #9 due to satellite/issues).
- *g, r, i* bands: 11 frames each.

- **Dark Frames:**

- *Dark Flats*: 51 frames taken at 5s and 4s to match the flat-field exposures.
- *Dark Lights*: 21 frames of 120s taken at 01:52 am, matching the scientific exposure time.

- **Scientific Exposures:** The target, NGC 300, was observed starting at 21:56. The plan included exposures of 120s in 1×1 binning. The approximate valid counts per filter were:

- *u*: 9 images.
- *g, r, i*: 32 images each.

Focus adjustments were performed multiple times during the night (at 21:23 and 00:28) using series of 3s exposures to counteract the temperature drop.

B. Sensor Characterization: Bias and Readout Noise

Before performing the standard reduction, a statistical analysis of the CCD sensor's noise properties was conducted using the set of 101 bias frames acquired. This step determine the intrinsic Readout Noise (RON) and to select the optimal stacking strategy (Mean vs. Median).

1. Bias Level and Histogram Analysis

The analysis of a single bias frame reveals the electronic offset (pedestal) added by the camera to avoid negative counts during the Analog-to-Digital conversion. As shown in the histogram generated by the analysis code (Fig. 1), the distribution of pixel values follows a Gaussian (normal) distribution.

The mean bias level was calculated at 980.53 ADU. The standard deviation of a single frame (σ_{single}) was found to be 9.82 ADU. The symmetry of the histogram suggests that the noise is dominated by random electronic fluctuations rather than structural artifacts.

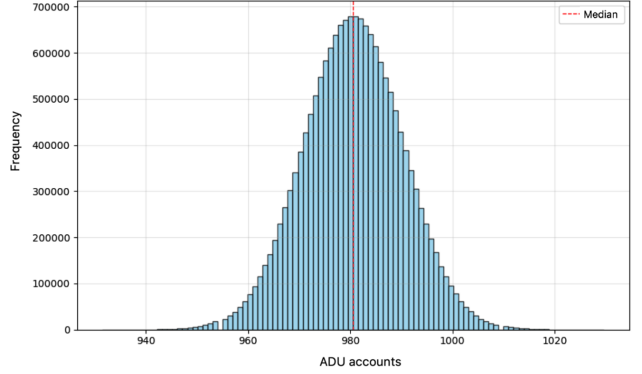


FIG. 1. Histogram of a single Bias frame showing a Gaussian distribution centered at the electronic offset of ~ 980 ADU.

2. Readout Noise (RON) Calculation

The standard deviation of a single bias frame includes both the random readout noise and the Fixed Pattern Noise (FPN). To isolate the intrinsic Readout Noise (RON), we applied the differential method. By subtracting two consecutive bias frames (B_1 and B_2), the FPN is subtracted out, leaving only the random noise components combined in quadrature. The RON is derived using the following relation:

$$\sigma_{diff}^2 = \sigma_{RON}^2 + \sigma_{RON}^2 \implies \sigma_{RON} = \frac{\sigma_{(B_1 - B_2)}}{\sqrt{2}} \quad (1)$$

Using 10 pairs of bias frames, our code calculated a mean RON of 9.7987 ADU. The fact that $\sigma_{RON} \approx \sigma_{single}$ (9.79 vs 9.82 ADU) indicates that the FLI ProLine 16801 camera has negligible Fixed Pattern Noise, and the signal is dominated by white noise.

3. Stacking Strategy and Residual Noise

To create the Master Bias, we evaluated two combination methods: Mean and Median. Theoretical statistics predict that for N uncorrelated images, the noise should decrease by a factor of $\sqrt{N}$:

$$\sigma_{Master} \approx \frac{\sigma_{single}}{\sqrt{N}} \quad (2)$$

For our dataset of $N = 101$ frames, the expected theoretical noise is $\sigma_{theo} = 9.82/\sqrt{101} \approx 0.97$ ADU. We compared the residual noise of the generated masters:

- **Master Mean:** $\sigma = 1.30$ ADU.
- **Master Median:** $\sigma = 1.42$ ADU.

The results show that the Master Mean achieves a lower noise level than the Median. This is consistent with statistical theory, which states that the median estimator is approximately $1.25\times$ noisier than the mean for Gaussian distributions. Since bias frames are short exposures (0s) and are not affected by cosmic rays, the outlier rejection provided by the median is unnecessary. Therefore, the Mean combination was selected for the final reduction pipeline.

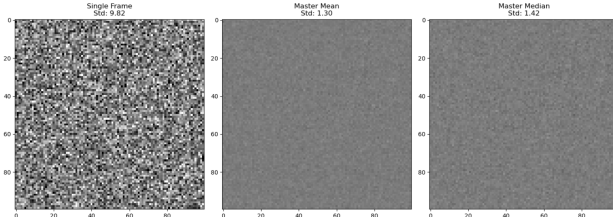


FIG. 2. Comparison of residual noise. Left: Single frame. Center: Master Mean. Right: Master Median. The Mean combination provides the smoothest background.

C. Dark Current Characterization and Gain Estimation

Following the bias analysis, we characterized the thermal noise (Dark Current) of the sensor and estimated the conversion gain of the CCD. These parameters are essential for correct image calibration and flux conversion.

1. Comparison of Master Darks and Master Bias

Dark frames contain both the readout offset (Bias) and the thermal signal accumulated during the exposure time. To isolate the thermal component, we compared the Master Darks with the Master Bias.

a. Long Exposures (120s): For the scientific exposure time (120s), the mean level of the Master Dark was found to be 984.04 ADU. Compared to the Master Bias level (979.83 ADU), this yields a net thermal signal of:

$$\Delta_{120s} = \text{Mean}_{\text{Dark}120} - \text{Mean}_{\text{Bias}} = 4.21 \text{ ADU} \quad (3)$$

While the mean thermal accumulation is low (~ 0.035 ADU/s), the spatial analysis (Fig. 3, right panel) reveals significant non-uniformities, specifically bright “Hot Pixels” and a noticeable glow in the upper-right corner, known as Amp Glow. This confirms that simple Bias subtraction is insufficient for science frames; a proper Master Dark is required to subtract these localized thermal artifacts.

b. Short Exposures (5s - Dark Flats): We also compared the Master Dark Flat (5s) against the Master Bias. The difference was found to be 1.91 ADU although the difference is small compared to the 120s frames, it is non-zero. The Master Dark Flat is technically better for calibrating flat fields than the Master Bias. This is because it accounts for the small but systematic thermal accumulation during the 4-5 seconds of flat exposure. Using only Bias would leave a residual of ~ 2 ADU, potentially introducing a systematic error in the flat-field normalization.

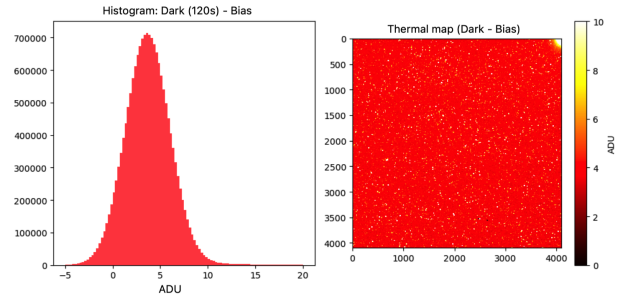


FIG. 3. Dark frame analysis. On the left: Thermal signal histogram (dark skew). On the right: Thermal map showing hot pixels and amplifier glow in the upper right corner.

2. CCD Gain Estimation

The Gain factor (g , in e^-/ADU) converts the digital counts into physical electrons. We estimated this parameter using the photon transfer method on the central region of the illuminated Flat frames (filter g).

To isolate the photon shot noise (σ_{shot}) from the fixed sensitivity variations (flat-field structure), we used the difference of two identical flat frames (F_1, F_2) corrected by the Master Bias:

$$\sigma_{\text{shot}}^2(\text{ADU}) = \frac{\text{Var}(F_1 - F_2)}{2} \quad (4)$$

Our code calculated a shot noise variance of 21,372.61 ADU^2 for a mean signal level of 39,145.93 ADU. Assuming Poisson statistics where the variance in electrons equals the mean signal in electrons ($\sigma_{e^-}^2 = N_{e^-}$), the Gain is derived as:

$$g = \frac{\text{Mean Signal (ADU)}}{\text{Variance}_{\text{shot}} (\text{ADU})} \quad (5)$$

Substituting our measured values:

$$g = \frac{39145.93}{21372.61} \approx 1.832 \text{ } e^-/\text{ADU} \quad (6)$$

This value indicates that the FLI ProLine camera generates 1 ADU for approximately every 1.83 photo-electrons collected, a value well within the expected range for this instrument.

D. Flat Field Characterization and Normalization Strategy

The flat-field correction is intended to map the pixel-to-pixel sensitivity variations ($S_{x,y}$) and the optical transmittance ($T_{x,y}$) of the system. However, the light source used for these calibrations (twilight sky) is not constant in time. For this reason, we analyzed the stability of the acquired frames to determine the correct stacking strategy.

1. Stability Analysis

We measured the mean counts in a central region (200×200 pixels) for every flat frame across all filters (u, g, r, i). As illustrated in Fig. 4, the intensity of the source shows a steep, linear decline characteristic of evening twilight observations.

The analysis reveals significant variations in intensity during the acquisition sequence:

- **Filter u :** 34.38% variation.
- **Filter g :** 34.03% variation.
- **Filter r :** 37.80% variation.
- **Filter i :** 47.29% variation.

This drastic change in illumination indicates that a simple average combination is not suitable, as the earlier (brighter) frames would dominate the signal-to-noise ratio and weight of the final master frame, potentially biasing the result.

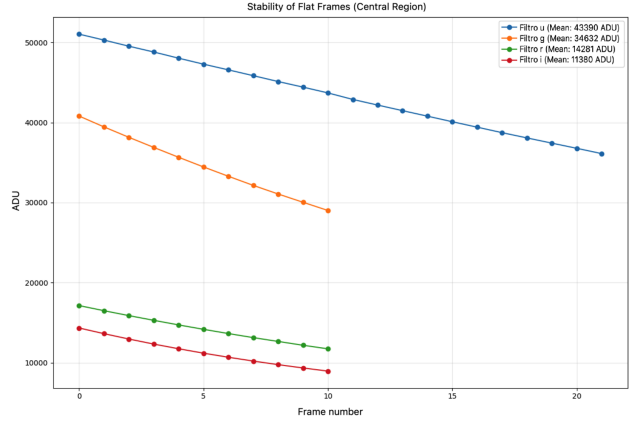


FIG. 4. Stability analysis of Flat frames showing the mean count decay in the central region for all filters. The decreasing trend confirms the dimming of the twilight sky during acquisition.

2. Normalization Strategy

Given the non-stability of the source, we adopted an individual Normalization Strategy. Before combining the images, each individual flat frame F_i was corrected for the dark current (using the Master Dark Flat) and then normalized by its own mean value μ_i .

The mathematical operation for each frame i is:

$$F'_i(x, y) = \frac{F_i(x, y) - \text{MasterDarkFlat}(x, y)}{\mu_i} \quad (7)$$

where $\mu_i = \langle F_i - \text{MasterDarkFlat} \rangle$.

This ensures that the mean value of every frame is exactly 1.0, effectively removing the temporal variation of the lamp/sky brightness while preserving the spatial sensitivity structure. Finally, the Master Flat was created by averaging these normalized frames:

$$\text{MasterFlat}(x, y) = \frac{1}{N} \sum_{i=1}^N F'_i(x, y) \quad (8)$$

3. Photometric Effects of Non-Flatness

The non-flatness of the image represents the combined effect of vignetting (optical fall-off towards the corners) and Quantum Efficiency (QE) variations among pixels. The observed image I_{obs} relates to the true flux I_{true} by:

$$I_{obs}(x, y) \approx I_{true}(x, y) \cdot T(x, y) \cdot S(x, y) \quad (9)$$

If this effect is not corrected (or poorly corrected), the photometric measurements become spatially

dependent. A star located in the vignetted corners would appear systematically fainter than an identical star in the center of the CCD. Furthermore, noise in the Master Flat propagates to the scientific image according to:

$$\sigma_{cal}^2 \approx \sigma_I^2 + \left(\frac{I}{F}\right)^2 \sigma_F^2 \quad (10)$$

Therefore, ensuring a high signal-to-noise ratio in the Master Flat is essential to prevent degrading the quality of the scientific data, particularly in the bright regions as we will see in the next section for image processing.

E. Image Reduction and Reconstruction Pipeline

Following the characterization of the sensor and the derivation of the master calibration frames, the scientific images of NGC 300 were processed using a custom Python pipeline designed to address specific artifacts identified during the analysis phase for M83 in Astronomia Observacional I.

1. Individual Calibration and Artifact Removal

Each of the raw scientific exposures (I_{raw}) was calibrated using the standard linear reduction equation, but with a critical modification to handle the "Hot Pixels" artifacts. The basic calibration is given by:

$$I_{cal}(x, y) = \frac{I_{raw}(x, y) - \text{MasterDark}(x, y)}{\text{MasterFlat}_{norm}(x, y)} \quad (11)$$

During the initial processing tests, we identified significant "black hole" artifacts (pixels with negative values) caused by the subtraction of saturated hot pixels in the Master Dark. These artifacts, often referred to as "black worms" after alignment, propagate errors during the deconvolution step. To mitigate this, an automated cleaning step was integrated into the `reduce_light` function and sigma clipping:

$$I'_{cal}(x, y) = \begin{cases} \text{median}(I_{cal}) & \text{if } I_{cal}(x, y) < 0 \\ I_{cal}(x, y) & \text{otherwise} \end{cases} \quad (12)$$

This thresholding strategy effectively replaces physically impossible negative values with the local background level, preventing the formation of dark artifacts in the final stack.

2. Intra-Filter Alignment and Stacking

Due to the lack of guiding and the wind turbulence during the night, significant shifts occurred between individual exposures. To correct this, we employed the `phase_cross_correlation` algorithm from the `skimage.registration` package. This method calculates the translation ($\Delta x, \Delta y$) between a reference frame and each target frame in the Fourier domain, achieving sub-pixel accuracy.

Once aligned, the images for each filter were combined. Based on our sensor analysis, we switched from a Mean combination (optimal for Bias) to a median combination for the scientific lights. The median is robust against transient events, such as cosmic rays and the white worm trails left by hot pixels that shift due to dithering/drift:

$$I_{stack}(x, y) = \text{median}(I'_{cal,1}(x, y), \dots, I'_{cal,N}(x, y)) \quad (13)$$

3. Inter-Filter Registration

Finally, to produce a color composite, the four stacked master images (u, g, r, i) needed to be aligned to a common reference frame. We selected the r -band stack as the geometric reference due to its high signal-to-noise ratio. The remaining filters were registered using the χ^2 shift minimization algorithm (`image_registration.chi2_shift`), which is robust against noise differences between spectral bands.

The final result is a set of four deep, artifact-free, and perfectly registered images ready for deconvolution and RGB synthesis.

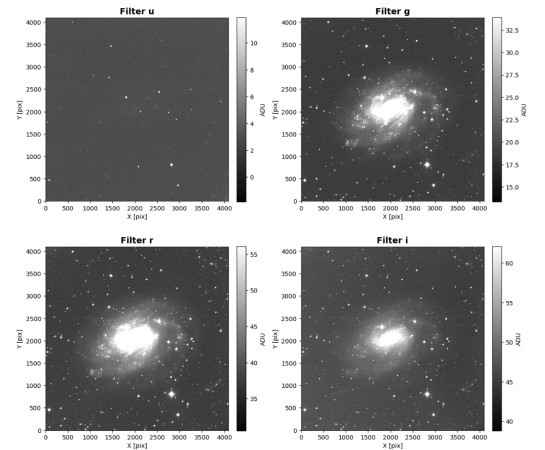


FIG. 5. Final stacked and aligned images of NGC 300 in the u, g, r, i bands after the full reduction pipeline. Note the removal of hot pixel artifacts and the smooth background achieved by stacking.

F. Deconvolution and Resolution Enhancement

The final stage of the processing pipeline aimed to partially recover the spatial resolution lost due to atmospheric turbulence (seeing) and optical diffraction. This degradation can be modeled mathematically as a convolution operation. If I_{obs} is the observed image, I_{true} is the true light distribution, and P is the Point Spread Function (PSF), the image formation process is described by:

$$I_{obs}(\vec{x}) = (I_{true} \otimes P)(\vec{x}) + N(\vec{x}) \quad (14)$$

where $\otimes$ denotes the convolution operator and $N(\vec{x})$ represents the additive noise. Deconvolution is the inverse problem of estimating I_{true} given I_{obs} and an estimate of P .

1. Richardson-Lucy Algorithm

For this work, we implemented the Richardson-Lucy (RL) deconvolution algorithm. This is an iterative method based on Bayesian statistics that seeks to maximize the likelihood of the restored image assuming Poisson noise statistics.

Starting with an initial estimate $u_0 = I_{obs}$, the algorithm updates the estimate u_k at iteration k using the following multiplicative rule:

$$u_{k+1} = u_k \cdot \left(\frac{I_{obs}}{u_k \otimes P} \otimes P^* \right) \quad (15)$$

Here, P^* represents the adjoint (or mirrored) PSF. The term in the denominator ($u_k \otimes P$) represents the re-blurred current estimate, which is compared to the observed image.

2. Implementation and Results

The algorithm was implemented in Python using `scipy.signal.fftconvolve` to accelerate the convolution steps in the Fourier domain. Since a high-SNR isolated star was not available to extract an empirical PSF, we generated a Synthetic Gaussian PSF with a standard deviation $\sigma = 2.0$ pixels, approximating the seeing conditions of the night.

We applied 15 iterations to the aligned stacks of NGC 300. The results show a significant improvement in the definition of the galactic nucleus and the star-forming regions in the spiral arms. However, as expected with the Richardson-Lucy method, the process also introduced specific artifacts:

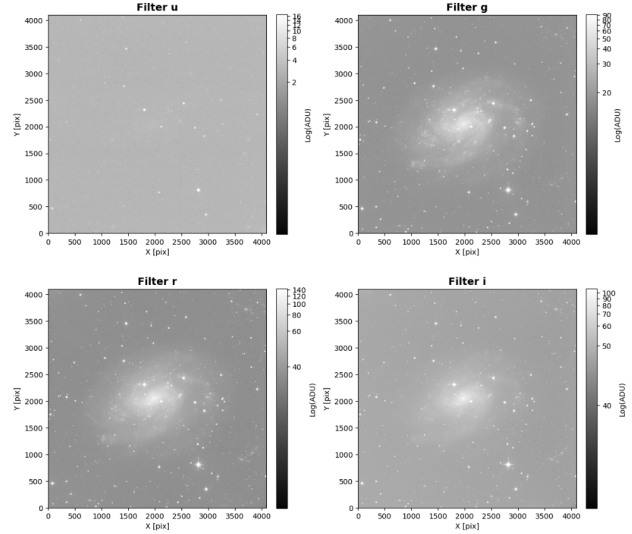


FIG. 6. Results obtained after deconvolution.

1. **Noise Amplification:** The background noise became more granular ("mottled") as the algorithm attempts to fit noise peaks as real sources.
2. **Ringing (Gibbs Phenomenon):** Dark rings appeared around the brightest saturated stars, a known side effect when the synthetic PSF does not perfectly match the true optical PSF.

For this reason, no further iterations were implemented. Despite these artifacts, the deconvolved images provided a superior level of detail for the morphological analysis of the galaxy and were removed during editing using Photoshop applications.

G. Post-Processing and RGB Composition

To visualize the photometric data as a coherent color image, a two-stage post-processing workflow was implemented using the ESA/ESO/NASA FITS Liberator and Affinity Photo (Canva) software.

1. Dynamic Range Compression

The deconvolved FITS images possess a high dynamic range (32-bit float) that cannot be directly displayed on standard monitors. To map these physical intensities into a visible 16-bit range, we utilized FITS Liberator.

For each filter (g, r, i), the black and white points were adjusted to clip the background noise and the saturation peak, respectively. The stretch function selected was the ArcSinh (Inverse Hyperbolic Sine). Unlike

a simple logarithmic stretch, the ArcSinh function behaves linearly for low pixel values (preserving the faint background structure of the galaxy arms) and logarithmically for high values (preventing the saturation of bright stars). The transformation is described by:

$$f(x) = \text{asinh}(\alpha \cdot x) \quad (16)$$

where α is a scaling factor determining the nonlinearity of the stretch. The processed channels were exported as 16-bit TIFF files.

2. Layer Composition and Color Mapping

The TIFF files were imported into Affinity Photo to assemble the final composite image. A layer-based architecture was used (Fig. 7), where the filters were mapped to chromatic channels according to their wavelength, creating a representative color image:

- **Red Channel (0°):** Assigned to the i filter (longest wavelength).
- **Green Channel (120°):** Assigned to the r filter.
- **Blue Channel (240°):** Assigned to the g filter.

a. Role of the u Filter: The u band image presented a unique challenge: due to the intrinsic spectral energy distribution of NGC 300, this filter captured very little flux from the diffuse galactic disk, containing mostly point sources (young massive stars and foreground stars). This characteristic was exploited to correct artifacts. The u layer was implemented as a Luminance Mask to recover the profiles of the most saturated stars, which had lost definition in the deeper g, r, i exposures.

3. Cosmetic Adjustments

Final adjustments were applied using non-destructive adjustment layers. Curves and Levels were tweaked to balance the color background, and a Selective-Color adjustment was used to separate the blue star-forming knots from the warmer galactic core. Finally, a specific Noise Reduction filter was applied to smooth the background grain amplified by the Richardson-Lucy deconvolution.



FIG. 7. Layer composition in Affinity Photo applied to the g, r, i, u channels.

III. CONCLUSION

The, successfully demonstrated the complete workflow of astronomical imaging, from data acquisition to scientific reduction and aesthetic reconstruction. Despite challenging weather conditions characterized by strong winds and technical interruptions, the acquired dataset allowed for a profound analysis of both the CCD performance and the galactic morphology.

The sensor characterization phase yielded critical physical parameters for the FLI ProLine 16801 camera. We determined a Readout Noise (RON) of ~ 9.8 ADU with a Gaussian distribution and an estimated Gain of $1.832 \text{ e}^-/\text{ADU}$. These values guided our reduction strategy, specifically the choice of Mean stacking for Bias frames (to minimize statistical noise) versus Median stacking for scientific exposures (to eliminate cosmic rays and transient artifacts). Furthermore, the

analysis of the Dark frames highlighted the necessity of correcting the Amp Glow present in the upper-right corner, which a simple Bias subtraction would have failed to remove.

A significant challenge was the instability of the twilight flat-fields, which exhibited an intensity drop of up to 47% during acquisition. This was effectively mitigated by implementing an individual normalization strategy within our custom Python pipeline, ensuring a mathematically correct flat-field correction. Additionally, the pipeline successfully addressed specific artifacts such as the black worms (negative pixel values) and white worms (moving hot pixels) through thresholding and median combination, respectively.

The application of the Richardson-Lucy

deconvolution algorithm proved to be a powerful tool for recovering spatial resolution. Although it introduced a trade-off by slightly amplifying the background noise and creating minor ringing around saturated stars, it significantly enhanced the definition of the galactic nucleus and the star-forming regions in the spiral arms.

Finally, the post-processing stage synthesized these scientific data into a coherent color image. By using the *u*-band data as a luminance mask and applying an Asinh stretch, we managed to preserve the dynamic range of the galaxy, revealing faint structures while controlling stellar saturation. All these results are located in the drive proposed for observation and the repository with the implemented codes and their requirements.

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